

COSC 304 Documentation

Executive Summary

This project involves developing a dynamic e-commerce web application for a grocery store, allowing customers to browse products, manage their shopping carts, and place orders. The system also incorporates admin functionality for managing users, viewing orders, and overseeing customer activities. The platform uses a backend powered by a SQL Server database, and the frontend is designed with a responsive layout for seamless user interaction.

Project Objectives

1. Customer-Centric Features:

- Customers can browse and view detailed product information.
- They can add items to their shopping cart, review orders, and make purchases.
- A review system allows customers to rate and comment on products, but it is limited to one review per product per customer.

2. Admin Functionality:

- Admin users have exclusive access to manage customer information, review all orders, and oversee platform activity.
- Admin privileges include creating new users, editing customer information, and viewing all registered users.

3. Data Integrity and Security:

- Authentication ensures that only logged-in users can access personalized features.
- Admin privileges are verified before granting access to sensitive operations.
- Data validation is implemented across all forms to maintain data consistency and security.

4. Scalability and Maintainability:

- The database schema is designed with normalized tables and relationships to support future scalability.
 - The application architecture uses reusable components like headers, footers, and templates for efficient maintenance.
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Key Features

1. Product Display:

- Products are dynamically fetched from the database and displayed with images, descriptions, and prices.
- Customers can continue shopping or proceed directly to checkout from the product page.

2. Review System:

- Customers can view existing reviews and add their own feedback if they haven't reviewed the product already.
- Reviews include a rating (1-5) and a comment.

3. User Management:

- Admins can create new user accounts, update existing user details, and view all registered users.

4. Order History:

- Customers can view their order history filtered by their username.
- Admins can view a comprehensive list of all orders.

5. Responsive Design:

- The platform includes a consistent header and footer design for improved navigation and branding.
 - The layout is responsive, ensuring usability across devices.
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Technology Stack

- **Frontend:** HTML, CSS, JavaScript, and JSP for rendering dynamic content.
 - **Backend:** Java with JDBC for database interaction.
 - **Database:** SQL Server with a well-structured schema to handle customers, orders, reviews, products, and categories.
 - **Web Server:** Tomcat.
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Challenges Addressed

1. **Data Validation:** All input fields are validated to prevent invalid data, ensuring smooth application functionality and data accuracy.
 2. **Authentication:** Secure user authentication mechanisms prevent unauthorized access.
 3. **Concurrency:** The system supports multiple users interacting simultaneously without data conflicts.
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Future Scope

1. **Enhanced Features:**
 - Implement advanced search and filtering options for products.
 - Add support for multiple payment gateways.
 2. **Performance Optimization:**
 - Improve query performance and caching for faster page loading.
 - Use load balancers and scalable infrastructure for increased user traffic.
 3. **Mobile Application:**
 - Develop a mobile app version to enhance customer accessibility.
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This project represents a robust e-commerce platform tailored to meet the needs of both customers and administrators, ensuring usability, security, and scalability. By

leveraging modern web technologies and a well-structured backend, the system is designed to provide a seamless shopping experience while enabling efficient management for administrators.

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System Features

The grocery store web application is designed with a range of features aimed at ensuring a seamless shopping experience for customers while providing robust administrative capabilities for managing the platform. Below is a detailed breakdown of the system features:

Customer Features

1. Product Browsing:

- Customers can view all available products with detailed information, including:
 - Product name
 - Price
 - Description
 - Images
- A responsive design ensures smooth navigation across devices.

2. Shopping Cart:

- Add products to the cart with the ability to:
 - View selected items.
 - Modify quantities or remove items.
- Dynamic updates to the total cart value.

3. Order Placement:

- Customers can place orders by proceeding to checkout directly from their cart.
- Shipping address fields are pre-filled with customer details for convenience.

4. **Order History:**

- Logged-in customers can view a personalized list of their past orders.
- Each order displays the order date, items purchased, quantities, and total cost.

5. **Product Reviews:**

- Customers can read reviews left by other users.
- They can leave their own review for a product, including:
 - A rating (1-5).
 - A comment about their experience.
- Users are restricted to one review per product, ensuring fair and unbiased feedback.

6. **Secure User Authentication:**

- User login ensures access to personalized features.
 - Customer details are securely stored and retrieved using encrypted sessions.
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Admin Features

1. **User Management:**

- Admins can:
 - View a list of all registered users.
 - Add new users with roles (e.g., customer or admin).

2. **Order Management:**

- Access a full list of all orders placed on the platform.

3. **Access Control:**

- Admin-exclusive pages are protected to ensure only authorized users with admin privileges can access them.
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Shared Features

1. Header and Footer:

- A consistent and user-friendly design:
 - The header includes navigation links to key pages (e.g., Home, Products, Orders, Cart).
 - Admin-only options (e.g., "Admin Panel" and "Create User") are displayed for admin users.

2. Data Validation:

- All input fields (e.g., forms for reviews, user creation, and order placement) are validated to prevent:
 - Invalid data submissions (e.g., numbers in name fields, invalid email formats).
 - Malicious inputs like SQL injections.
-

System Integration

1. Database Features:

- A SQL Server database is used to manage:
 - Product catalog and categories.
 - Customer information.
 - Orders, reviews, and payment details.
- Relationships between tables are normalized for data integrity.

2. Dynamic Content Rendering:

- JSP pages dynamically fetch and display content from the database.
- Customers and admins see personalized content based on their roles and actions.

3. Error Handling:

- Clear and user-friendly error messages for invalid inputs or unauthorized access.
 - Logging of errors for admin debugging purposes.
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Security Features

1. Authentication and Authorization:

- Role-based access ensures only admins can access critical pages like user management and order overviews.
- Logged-in users are identified via secure session tokens.

2. Input Sanitization:

- All user input is sanitized to prevent SQL injection and other common vulnerabilities.

3. Encrypted Credentials:

- Passwords are securely stored in the database to protect user data.
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Web Sources and AI Used

The development of the grocery store web application involved utilizing a combination of technical resources, documentation, and best practices from trusted web and AI sources. Below is a categorized list of the resources referenced during this project:

Web Development Sources

1. JSP and Java Resources:

- Oracle Java EE Documentation: Official documentation for Java EE technologies, including JSP and Servlets, used for dynamic content rendering and session management.
 - <https://docs.oracle.com/javaee/>
- JDBC Documentation: Guide for establishing database connections, executing SQL queries, and managing transactions.
 - <https://docs.oracle.com/javase/8/docs/technotes/guides/jdbc/>

2. HTML, CSS, and UI Design Resources:

- Mozilla Developer Network (MDN): Comprehensive tutorials and references for HTML, CSS, and responsive design techniques.
 - <https://developer.mozilla.org/>

- Bootstrap Documentation: Resources for creating responsive and mobile-friendly web designs.
 - <https://getbootstrap.com/>
-

AI Resources

1. Natural Language Assistance:

- OpenAI's GPT-4 model was utilized for generating, refining, and validating code, documentation, and recommendations throughout the project.
 - <https://openai.com/>
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Walkthrough for the Project: Grocery Store Website

1. Login and Authentication

Currently, there is only one admin user whose credentials are -

Username - arnold

Password - 304Arnold!

1. Login Page:

- Customers and admins can log in through a common page (`login.jsp`).
- The session is used to store authenticated user details.
- Admin users are identified using the `isAdmin` flag from the `customer` table.

Your Supermart

Home Products Order List Cart Edit User Orders Login

Please Login to System

Username

Password

Log In

About Us
Your Supermart is dedicated to providing quality products at unbeatable prices. Shop with us for an exceptional experience!

Quick Links
Home
Products
Orders
Contact Us

Contact Us
+1-800-555-1234
support@yoursupermart.com
123 Supermart Lane, Cityville, USA

Follow Us

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2. Role-Based Access:

- If the user is an admin, they gain access to additional features like the Admin Dashboard.
- Non-admin users are restricted from accessing admin functionalities.

Your Supermart

HomeProductsOrder ListCartEdit UserOrders

Login

Please Login to System

You do not have admin privileges to access the URL http://localhost/shop/admin.jsp

Username:

Password:

Log In

About Us

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Quick Links

HomeProductsOrdersContact Us

Contact Us

+1-800-555-1234
support@yoursupermart.com
123 Supermart Lane, Cityville, USA

Follow Us

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2. Browsing Products

1. Product Listing Page (`listprod.jsp`):

- Customers can search for products by name or category.
- Products are displayed in a grid using Bootstrap cards for consistent styling.

Your Supermart

HomeProductsOrder ListCartEdit UserOrders

Login

Search for the products you want to buy:

Product Name:

Enter product name

Category:

-- Select a Category --

SearchReset

Product List

Chai

leverages

0 boxes x 20 bags

18.00

add to cart

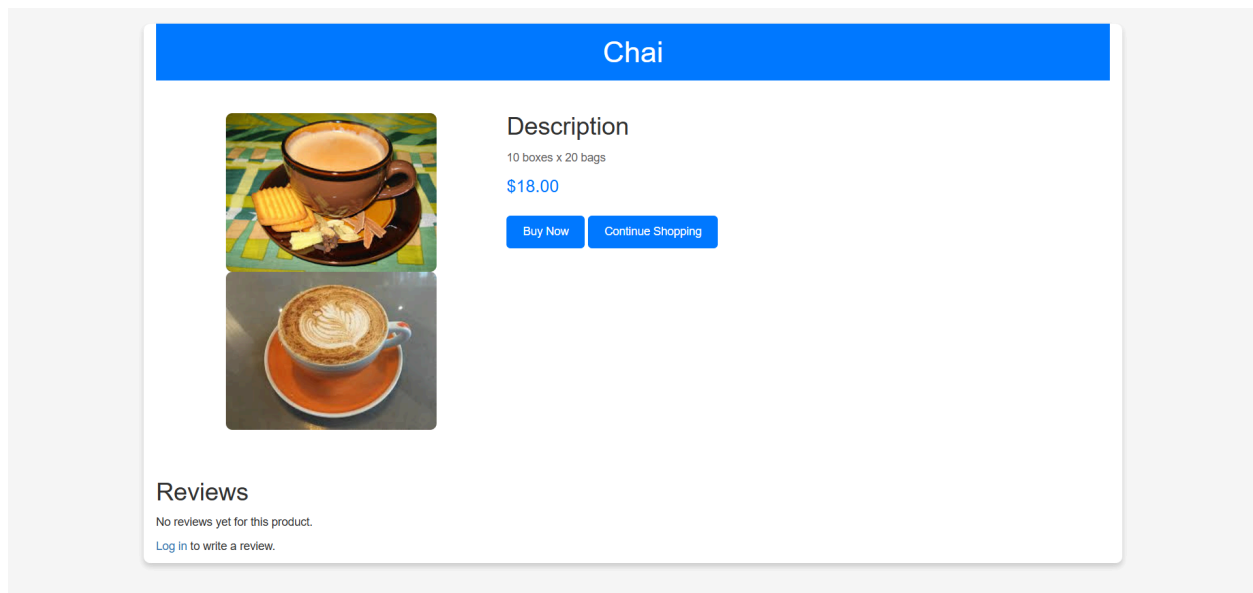
Chang

leverages

4 - 12 oz bottles

2. Individual Product Details:

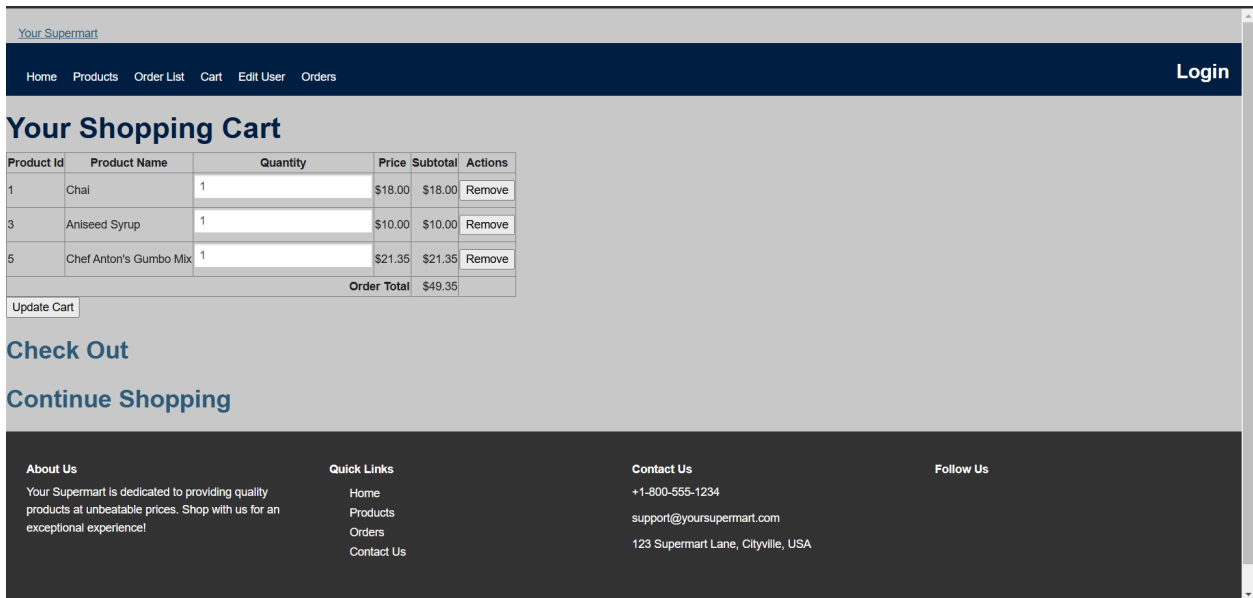
- Each product card links to a detailed product card as a temporary page (`product.jsp`) with a description, price, and an option to add it to the cart.



3. Shopping Cart

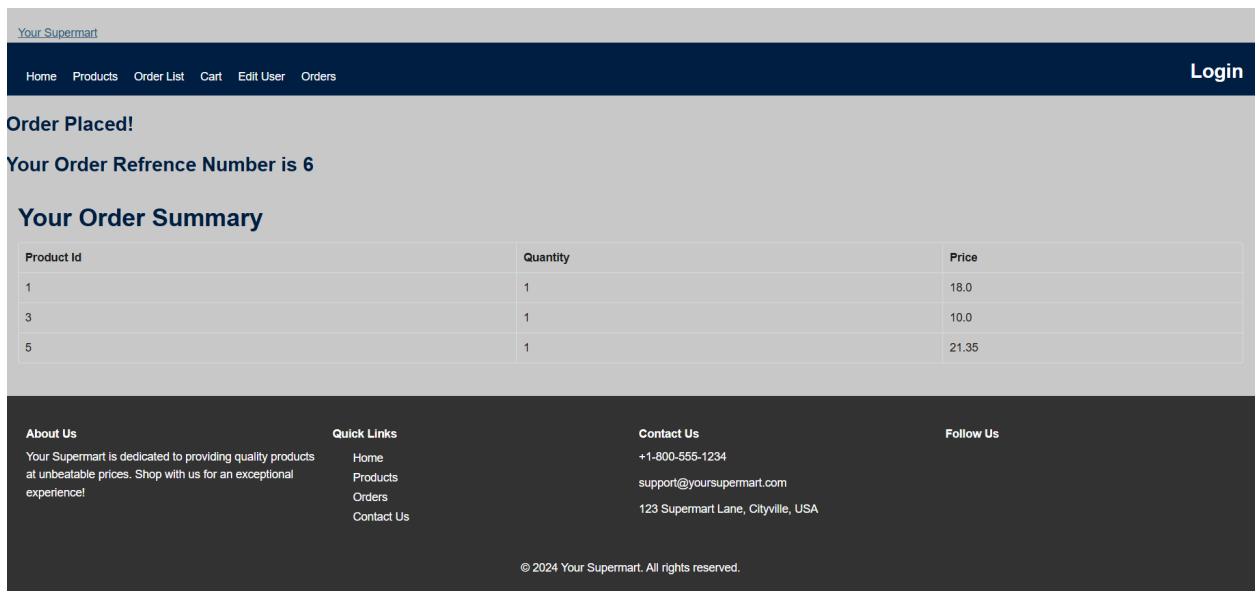
1. Cart Management (`showcart.jsp`):

- Users can add products to their cart.
- The page displays all selected items with their names, quantities, and total price.
- Users can update quantities or remove items.



2. Checkout Process:

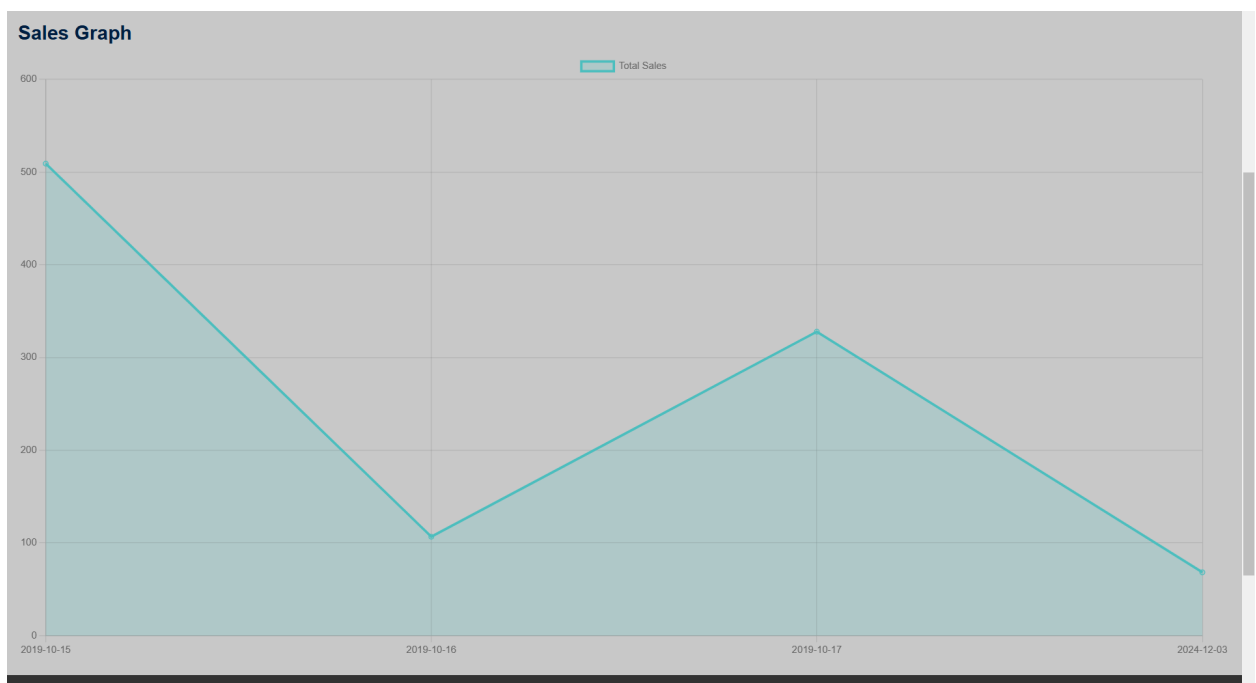
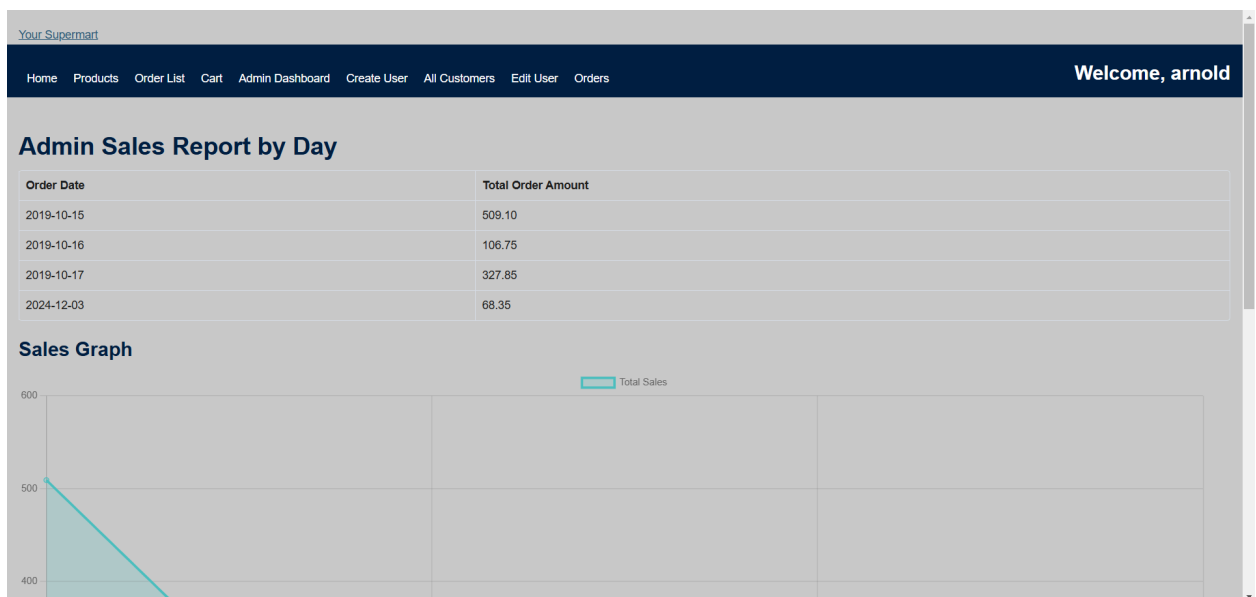
- On proceeding to checkout, the order is saved in the **ordersummary** table with the total amount and order date.



4. Admin Features

1. Admin Dashboard (`admin.jsp`):

- A central page for admins to manage the store.
- Links to features like user management, and viewing sales reports.



2. Viewing All Customers (**listCustomer.jsp**):

- Displays a table of all customers with details like names, email and User Id.

Your Supermart

Home Products Order List Cart Admin Dashboard Create User All Customers Edit User Orders **Welcome, arnold**

All Users

Customer ID	First Name	Last Name	Email	Phone Number	City	Country	User ID	Is Admin
1	Arnold	Anderson	a.anderson@gmail.com	204-111-2222	Winnipeg	Canada	arnold	Yes
2	Bobby	Brown	bobby.brown@hotmail.ca	572-342-8911	Boston	United States	bobby	No
3	Candace	Cole	cole@charity.org	333-444-5555	Chicago	United States	candace	No
4	Darren	Doe	oe@doe.com	250-807-2222	Kelowna	Canada	darren	No
5	Elizabeth	Elliott	engel@uiowa.edu	555-666-7777	Iowa City	United States	beth	No

[Back to Home](#)

About Us
Your Supermart is dedicated to providing quality products at unbeatable prices. Shop with us for an exceptional experience!

Quick Links
[Home](#)
[Products](#)
[Orders](#)
[Contact Us](#)

Contact Us
+1-800-555-1234
support@yoursupermart.com
123 Supermart Lane, Cityville, USA

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3. Create Users (**createUser.jsp**):

- Admins can create new users with details like names, email, and userId.

Home Products Order List Cart Admin Dashboard Create User All Customers Edit User Orders **Welcome, arnold**

Create a New User

First Name:

Last Name:

Email:

Phone Number:

Address:

City:

State:

Postal Code:

Country:

User ID:

Password:

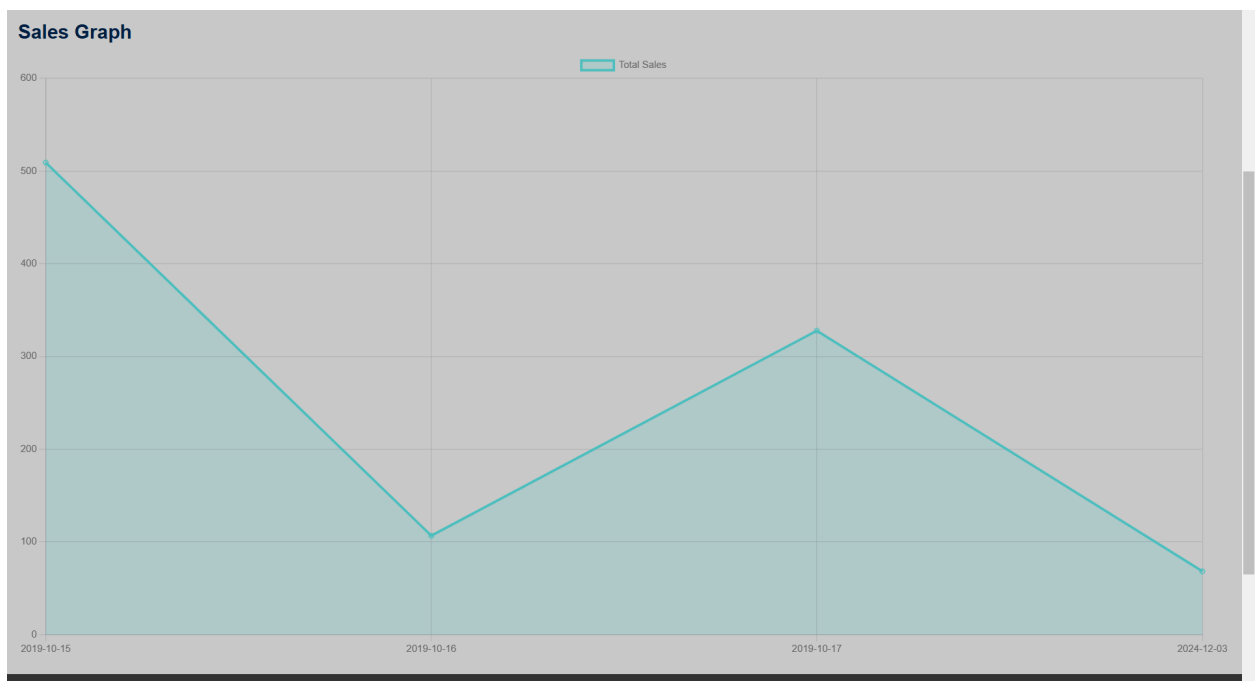
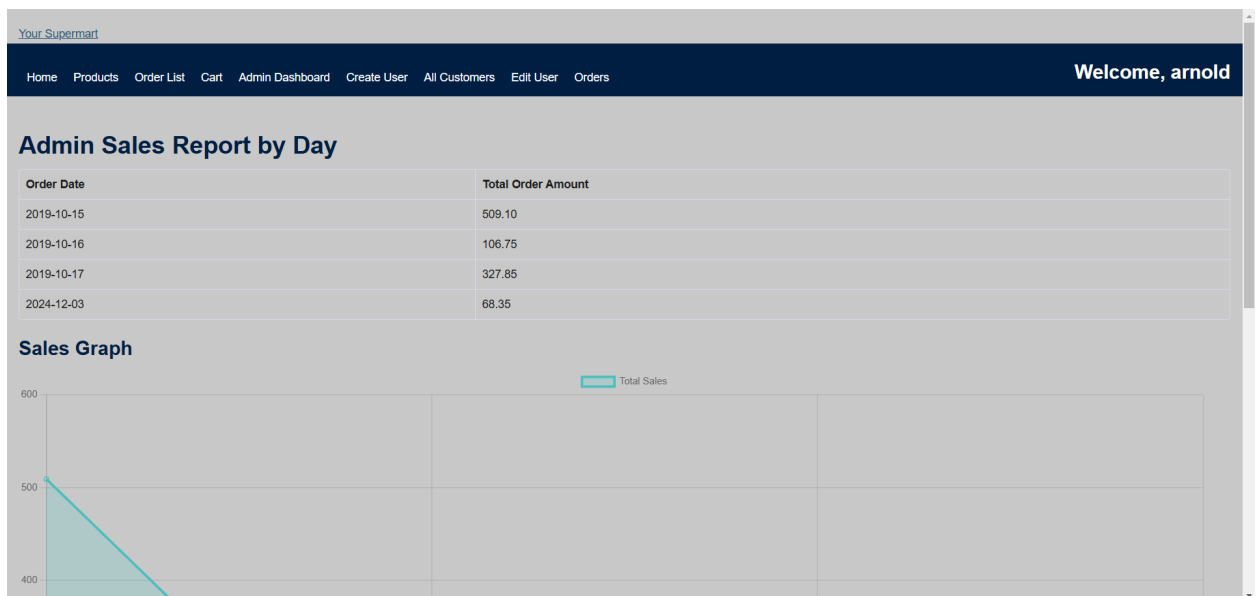
Is Admin: ☐ (Check to grant admin privileges)

[Back to Admin Dashboard](#)

5. Sales Reporting

1. Admin Sales Report by Day:

- Displays daily sales data in a tabular format, showing the order date and total sales amount.
- Includes a line graph (using Chart.js) for visualizing sales trends over time.



2. Graph Details:

- Dates are plotted on the x-axis, and total sales amounts are plotted on the y-axis.
 - Interactive features like tooltips allow users to view exact values for each day.
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Development Project Features List

Final Overall System Features	Category	Possible Points	Points	Comments / Code Files
Documentation (15 total points)				
Executive Summary (up to 1 page)	Core	2	2	304_Documentation.pdf
System Features (2 pages)	Core	2	2	304_Documentation.pdf
List of Web and AI sources used	Core	3	3	304_Documentation.pdf
Walkthrough with screenshots (up to 15 pages)	Core	8	8	304_Documentation.pdf
Hosting (10 total points)				
Running on cloud server	Advanced	10		
Main/Home Page (up to 14 total points)				
Search for a product by name	Core	1	1	listprod.jsp
Browse products by category	Core	1	1	listprod.jsp
List products (by search/browse)	Core	1	1	listprod.jsp
List products with image	Core	1	1	listprod.jsp
Page header with menu	Basic	1	1	listprod.jsp
Page header shows current logged in user	Basic	1	1	Header.jsp
Dynamic products on page based on sales	Basic	2		
Improved UI (user-friendly, images, etc.)	Basic	2	2	Card layout implimented using bootstrap to show products with animation
Product recommendation based on user	Advanced	4		
Shopping Cart (up to 8 total points)				
Add to shopping cart	Core	1	1	showcart.jsp
View shopping cart	Core	1	1	showcart.jsp

Update quantity (with data validation)	Basic	1	1	showcart.jsp
Remove item from shopping cart	Basic	1	1	showcart.jsp
Improved formatting/UI (e.g. in header bar)	Basic	1	1	Header.jsp
Cart stored in database between sessions	Basic	3		
Checkout (up to 9 total points)				
Checkout with customer id	Core	1	1	checkout.jsp
Checkout with payment/shipment info	Basic	2		
Checkout with data validation	Basic	2	2	checkout.jsp
Multiple shipments per order supported	Advanced	2		
Calculate taxes/shipping (by items/state)	Advanced	2		
Product Detail Page (up to 2 total points)				
Product detail page and item description	Core	1	1	product.jsp
Product detail has an image from database	Core	1	1	product.jsp
User Accounts/Login (up to 12 total points)				
Create user account page	Basic	2	2	createUser.jsp
Create account with data validation	Basic	2	2	processCreateUser.jsp
Edit user account info (address, password)	Basic	2	2	editUser.jsp
Login/logout	Core	1	1	login.jsp/logout.jsp
Register user/Forgot password email	Advanced	4		
Page listing all orders for user	Core	1	1	orders.jsp

Product Reviews (up to 5 total points)				
Ability to enter a review on a product	Basic	2	2	addReview.jsp
Display product review on product detail page	Basic	1	1	product.jsp
Restrict to one review per user on item purchased	Advanced	2	2	product.jsp
Warehouses/Inventory (up to 5 total points)				
Display item inventory by store/warehouse	Basic	2		
Edit item inventory by store/warehouse	Advanced	3		
Administrator Portal (up to 19 total points)				
Secured by login	Core	1	1	login.jsp
List all customers	Core	1	1	listCustomer.jsp
List report showing total sales/orders	Core	1	1	admin.jsp
Report with a graph	Advanced	3	3	admin.jsp
Add new product	Basic	2		
Update/delete product	Basic	2		
Change order status/ship order	Basic	1		
Upload a photo to file system for product	Advanced	2		
OR: Upload a photo to database for product	Advanced	4		
Database restore with SQL script	Basic	2	2	loaddata.jsp
Add/update warehouse, customer	Basic	2		
Database System/General				

Implement some validation using triggers	Advanced	2		
Use AJAX for some pages	Advanced	4		
User interface and navigation/usability		Up to 5	5	Implemented header and footer which eases navigation. Consistent Table design. Consistent Background. Used bootstrap to better the UI. eg: cards used to display products
Others (your suggestions)		Up to 5	5	Updated Customer table to have an isAdmin flag. That stores if a user is an admin. The header is dependent on the user, if admin then the header has all options. All admin features are restricted to admin and a non-admin is refused if trying to access admin features.
Total			64	